Contamination Robustness Values: Sensitivity Analysis for Benchmark Claims Under Unverifiable Data Leakage

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Abstract

Leaderboard statements of the form "model A beats model B" are read as measurements of capability, but they are observational estimates whose largest error source, leakage of test items into training data, is left out of the error model. The field has answered with detection, which needs a training corpus, a reference benchmark, or a reference model assumed clean, and which cannot settle the question a reader actually has: does the leakage change the conclusion? We adopt the answer causal inference reached for unmeasured confounding: stop detecting the bias and bound it. Contamination is treated as an identification failure. We define a Contamination Sensitivity Model, CSM±(Λ^+, Λ^-, π, Γ_sel, ε), in which every parameter has a measurable counterpart; derive sharp partial-identification bounds for the clean score; and define the contamination robustness value Γ^*, the smallest contamination strength that overturns a claim, computable from published scores alone. We show that the fully adversarial bound sits at the endpoint of a Rosenbaum-type selection family. The estimator Γ^* is consistent and $\sqrt{n}$-normal, and a paired bootstrap with Benjamini–Hochberg control certifies robustness across a claim corpus at a controlled false discovery rate. Experiments in which models are genuinely trained on leaked items confirm the assumed lift channel (envelope–headroom correlation ≥ 0.91 in 15 of 15 configurations) and expose two failure modes, two-sided lift and spillover, which we repair inside the model. A corpus-dilution experiment closes the calibration loop: a single diluted exposure lands inside the literature-derived range [0.10, 0.45], while repeated diluted exposures consolidate toward complete memorization, so calibration must be stratified by dose. A pre-registered audit of 58 adjacent-pair claims across 16 models and four Open LLM Leaderboard tasks finds 84.5% not contamination-robust (exact 95% CI [72.6%, 92.7%]; one-sided $p = 2.7\times10^{-21}$ against the pre-registered 25% threshold), with only 8 claims, all with margins above 5 points, earning false-discovery-controlled certificates; sweeping the frozen constant across its plausible range leaves the conclusion intact, and robustness proves to be a monotone function of rank distance. In the single-draw binary regime the rule provably reduces to a calibrated margin threshold, and no procedure can do better on such data. Sensitivity analysis turns an unanswerable question into an answerable one, with a one-number reporting standard any leaderboard can adopt.

Keywords—benchmark contamination; partial identification; sensitivity analysis; machine learning evaluation; leaderboards; Rosenbaum bounds; false discovery rate; pre-registration

1. Introduction

Every empirical claim in machine learning eventually rests on a benchmark score, and every benchmark score rests on an assumption nobody can check: that the test items never entered training. Items become benchmarks because they are public, and public items are scraped. The same popularity that makes an item worth benchmarking makes it likely to appear somewhere in a web-scale corpus, and a model that has seen an item answers it for reasons unrelated to the skill the item was written to measure.

The size of the resulting error is not speculative. Developers who ran their own contamination audits report large gaps between clean and dirty subsets: 15.3 points on HellaSwag for LLaMA-70B [1], and estimated gains of up to 14 points on flagged subsets in the Llama 3 report [2]. Independent clean mirrors of GSM8K recover drops of about 13 points. These are not rounding errors on a leaderboard whose top ten models sit within one or two points of each other. When a benchmark's dominant error term is an order of magnitude larger than the margins it adjudicates, the ordering it produces is being read past its resolution.

The field's response has been detection: n-gram overlap tests, membership-inference attacks, kernel-divergence estimators, and lineage audits [3]–[5]. Detection is valuable and we use its outputs, but it cannot carry the weight placed on it, because it fails structurally three times over. For a closed-weight model the pretraining corpus is unobservable, so detection is impossible in principle rather than merely inconvenient. It returns a contestable binary that vendors dispute and no protocol adjudicates. And it is silent on the only decision-relevant question: a benchmark can be heavily contaminated with its ranking intact, or barely contaminated with its ranking reversed, and a detector cannot tell those apart.

The statistical wing of evaluation reform has the opposite problem: it quantifies uncertainty carefully, but the wrong uncertainty. Rank intervals for leaderboards [6] propagate sampling error, which shrinks as the test set grows. Contamination bias does not; a larger contaminated benchmark is a more precisely measured wrong number. An interval that excludes the dominant error source is not an honest interval, and the partial-identification tradition [7] says what to do instead: report the set of values consistent with the data and with every assumption one is willing to defend.

Causal inference faced this impasse four decades ago and resolved it by changing the question. When a confounder cannot be measured, one does not try to detect it; one reports how strong it would have to be to overturn the conclusion. That is what Rosenbaum's sensitivity parameter, the E-value, and omitted-variable-bias bounds all do [8]–[10]. This paper ports that machinery to machine-learning evaluation. The reduction is, once stated, close to obvious: contamination is unmeasured confounding between training-set membership and measured skill, so benchmark scores are partially identified. Nothing after the reduction is obvious. The port needs a new sensitivity model, because contamination acts through a bounded, headroom-proportional channel rather than an odds ratio; a new calibration strategy, because contamination strength is a physical quantity one can measure by training models on leaked data; and a new design theory, because benchmark builders can buy identification with freshness and repeated measurement.

Our objective is therefore narrow: give every published comparison a single number saying how much contamination it can absorb before it flips, computable from what leaderboards already publish. The contributions are a potential-outcomes formulation of benchmark scores under latent contamination (Section 3); the Contamination Sensitivity Model CSM±(Λ^+, Λ^-, π, Γ_sel, ε) and sharp partial identification of the clean score, with grouped, stratified, selection-bounded, and spillover-robust variants, every axiom having a measured counterpart (Section 4); the contamination robustness value Γ^*, an evaluation analogue of the E-value (Section 5); estimation theory covering consistency and $\sqrt{n}$-normality of Γ^*, the paired item bootstrap, finite-draw bias, and false-discovery-controlled certificates (Section 6); ground-truth validation with emergent contamination, which confirmed the channel shape, falsified strict monotonicity, and surfaced spillover (Sections 7 and 8); contamination-strength priors calibrated from published evidence (Section 9); two case studies and a pre-registered audit of the Open LLM Leaderboard (Section 10); and design prescriptions that price freshness and repeated draws in units of certified robustness (Section 11), with an open, one-command-reproducible library.

Section 2 places the work among contamination detection, leaderboard statistics, and causal sensitivity analysis. Section 3 sets up the estimand; Sections 4 to 6 develop the model, the robustness value, and the inference theory; Sections 7 to 9 cover experimental design, validation, and calibration. Section 10 applies the method to real benchmarks and reports the pre-registered audit, including what it reduces to and why that reduction is the result rather than a defect. Sections 11 to 15 give design prescriptions, discussion, limitations, conclusions, and reproducibility.

2. Related work

Three literatures meet in this paper, and each supplies something the other two lack: contamination measurement supplies the physical constants, leaderboard statistics supplies the inferential target, and causal sensitivity analysis supplies the machinery.

Contamination detection and quantification. Surveys catalogue detection methods and document how fragile their assumptions are [3], [11], [12]. Quantification matters more to us than detection does. Magar and Schwartz [13] introduced controlled injection and separated memorization from exploitation, establishing that the size of the effect, not merely its presence, is measurable. Dose-response studies at scale show that lift grows with model capacity and with the number of test-set replicas, and shrinks as the surrounding corpus grows [14]; post-training can partly reverse the lift and can also resurface it [15]. ConStat [16] is the strongest performance-only method available, comparing a primary benchmark against a reference benchmark relative to reference models. CapBencher [17] builds an overfitting alarm into new benchmarks, and inference-time decontamination [18] mitigates leakage at evaluation time for models one can run. The strengths are real and the weakness is common: none answers a claim-level question, and each needs something legacy claims about closed models cannot supply, whether reference models, reference benchmarks, weight access, or a redesigned benchmark. We treat this literature as calibration input rather than as competition, and use ConStat as the natural baseline and convergent-validity check. Auditing has begun to spread beyond language models, with pretraining-exposure audits proposed for time-series foundation models [4], [19], the setting of our first case study.

Leaderboard statistics. Rank intervals [6] give sampling-uncertainty-aware rankings and are the zero-contamination special case of our bounds (Corollary 2). Perturbation and social-choice analyses document leaderboard instability and characterise how hard a benchmark is to rig [20]–[22], but they perturb the data rather than model the bias, so they diagnose fragility without supplying an estimand. Psychometric and evaluation-efficiency work models items and their difficulty [23], [24], not contamination, and its item parameters are themselves estimated from possibly contaminated responses.

Causal sensitivity analysis. Rosenbaum bounds [8], distributional propensity-score approaches [25], the E-value [9], omitted-variable-bias frameworks [10], and extensions to general treatment types and time-varying settings [26], [27] supply the epistemology and much of the technique; none, to our knowledge, has been applied to benchmark evaluation. A parallel construct-validity programme argues that evaluation needs measurement theory and better reporting, including item-level data releases [28]–[30]. We agree, and supply one of the estimators that programme has lacked, along with a concrete reason to release item-level and repeated-draw data: it is what entitles a benchmark to sharper identification.

3. Setup: the clean score is partially identified

Fix a model and a benchmark with items $i = 1, \ldots, n$. Let $y^*_i \in [0,1]$ be the clean score for item i, meaning the counterfactual per-item score the model would have obtained had item i never entered its training data. Let $y_i \in [0,1]$ be the score actually observed, and $c_i \in \{0,1\}$ a latent indicator that item i entered training. The estimand is the clean benchmark score

$$\theta = (1/n) \Sigma_i y^*_i.$$

Leaderboards report the observed mean $\hat{\mu} = (1/n) \Sigma_i y_i$ and treat it as an estimate of θ. That substitution is valid only if $c \equiv 0$. Some c_i will equal one, but the damage comes from the correlation between c and y^*. Items popular enough to be benchmarked are popular enough to be scraped, and their popularity tracks the very skill the benchmark measures, so contaminated items are not a random subset. This is textbook confounding with an unobservable confounder, and it has the textbook consequence: θ is not point-identified. The honest report is the identified set, that is, the values of θ consistent with the data and with whatever assumptions the analyst will defend, rather than a point estimate whose error bar omits its largest term.

Nothing about this can be fixed by collecting more data. The width of the identified set depends on the assumed contamination process rather than on sampling noise, so a larger test set leaves it exactly where it was. What the width does depend on is the assumptions, and those have to be stated in units that can be measured somewhere, or the exercise degenerates into choosing a comfortable number. Section 4 states them and Sections 7 to 9 measure them.

4. The contamination sensitivity model

4.1 The one-sided core

A sensitivity model restricts how far the observed data can depart from the ideal. It should be weak enough to be credible, strong enough to be informative, and phrased so its parameters can be estimated from outside evidence. Under CSM(Λ, π):

$$\text{(A1) } 0 \le y_i - y^*_i \le c_i \cdot \Lambda \cdot (1 - y^*_i), \quad \text{(A2) } (1/n)\, \Sigma_i\, c_i \le \pi.$$

Axiom A1 says that leakage closes at most a fraction Λ of an item's remaining headroom. Headroom is the substantive choice here. An item the model already answers correctly cannot be improved by memorising it, whereas an item at chance level can be improved a great deal, so the natural scale for contamination lift is the distance to a perfect score rather than an absolute number of points. $\Lambda = 0$ means leakage is inert. $\Lambda = 1$ means a leaked item is answered perfectly. Both extremes occur in our experiments (Section 8), which is why we treat the parametrisation as measured rather than assumed. Axiom A2 caps the contaminated fraction at π. The two parameters separate what contamination does from how much of it there is, which matters because they are calibrated from different evidence: Λ from injection studies, π from detection studies and provenance metadata.

Proposition 1 gives the sharp identified set. For $\Lambda < 1$ define the floor $\ell_i(\Lambda) = \max\{0, (y_i - \Lambda)/(1 - \Lambda)\}$ and the per-item deflation capacity $d_i(\Lambda) = y_i - \ell_i(\Lambda)$, which is how far item i's observed score could be pushed down while remaining consistent with A1. Then $\theta \in [\hat{\mu} - B(\Lambda, \pi), \hat{\mu}]$, where $B(\Lambda, \pi)$ is the mean of the $\lceil \pi n \rceil$ largest capacities, and both endpoints are attained. This is a knapsack problem. An adversary with a budget of πn contaminable items spends it where scores can be explained away most cheaply. Sorting and summing the largest $\lceil \pi n \rceil$ capacities costs $O(n \log n)$, so the bound is computable for any benchmark that fits in memory.

Corollary 1 bounds the width of the identified set by $\pi\Lambda$ up to a rounding term, because the worst per-unit-budget bias is exactly Λ, at items scoring $y_i = \Lambda$. This gives a distribution-free sanity check: contaminating 10% of a benchmark through a channel of strength 0.4 cannot move the mean by more than four points, whatever the score distribution. Corollary 2 records that $B(0, \pi) = B(\Lambda, 0) = 0$, so the identified set collapses to the point estimate and sampling-only analyses such as rank intervals [6] are the $\Lambda = 0$ special case. Their intervals are not wrong; they are conditional on an assumption they do not state.

Figure 1 shows the identified set as Λ varies at three budgets, on data where the clean score is known by construction. The true clean score sits inside the band at the true parameter values, and the band closes onto the observed mean as $\Lambda \to 0$, which is what Corollary 2 predicts.

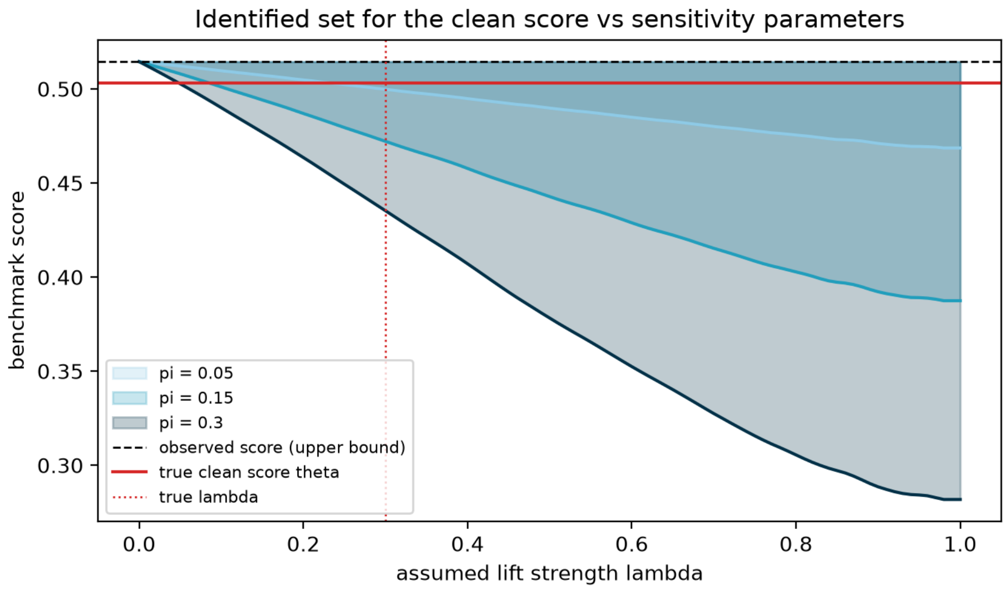

Figure 1: Sharp identified set for the clean score as the assumed lift strength Λ varies, at budgets $\pi \in \{0.05, 0.15, 0.3\}$, on synthetic data with known ground truth. The observed score (dashed) is always the upper endpoint, because one-sided leakage can only inflate. The true clean score θ (solid) lies inside the band at the true (Λ, π), and the set collapses to the observed mean as $\Lambda \to 0$, recovering sampling-only analysis.

4.2 Two measurement regimes

The sharpness of Proposition 1 depends on how informative the per-item scores are, and current benchmarks fall into two very different regimes. In regime R1 the scores are continuous, or binary but repeated, so the observed y_i take interior values and the capacities discriminate between items; the knapsack then binds strictly tighter than the distribution-free width.

Regime R2 is the single-draw binary case most language-model leaderboards publish: each item is scored 0 or 1 once. Here the machinery degenerates. Lemma A.1 shows $d_i(\Lambda) = 0$ for every item and every $\Lambda < 1$, with a jump to $d_i(1) = y_i$ at $\Lambda = 1$, so the sharp bound reports that only total memorisation could flip any claim. That is technically correct and practically useless. What remains honest in R2 is the population-level simple bound of width $\pi\Lambda$, in which Λ and π are identified only through their product. Our library detects the regime and refuses to report a sharp bound where it is vacuous. Section 11 prices the gap between regimes as a design lever.

4.3 Extensions forced by measurement

The core model needed four extensions, each of them forced by an experiment that falsified or strained it, and each measured in Section 8.

U1, two-sided lift. Strict monotonicity fails: leaked items are sometimes answered worse, through stale memorisation and through interference from post-training [15]. The model therefore admits a downward channel, $-\Lambda^- y^*_i \le y_i - y^*_i$, with $\Lambda^- = \rho\Lambda^+$. For a claim $A \succ B$ the worst case becomes A inflated and B deflated, which widens the region in which the claim can fail and makes every reported Γ^* more conservative.

U2, grouped contamination. Datasets leak whole rather than item by item, so the budget is spent on sources. The all-or-nothing group knapsack is bounded above by its fractional relaxation and is never wider than the item-level bound, so grouping can only help the analyst.

U3, provenance stratification. Items created after a model's training cutoff have $c_i = 0$ by construction, so any benchmark that records item dates hands the analyst a stratum of known-clean items and tightens the bound for free. This is the basis for the freshness prescription in Section 11.

U4, spillover. Training on leaked items can drift scores on clean items through format familiarity rather than answer memorisation. A bounded net drift ε widens both endpoints symmetrically, and is estimated from twin training runs rather than assumed; in our experiments ε ranges from 0.003 to 0.30 and grows with dose.

4.4 A bridge to Rosenbaum's family

The knapsack bound is adversarial: it assumes contamination lands exactly where it does the most damage, and a reviewer may object that real leakage is not maximally malicious. Proposition 3 answers by embedding the bound in a classical family. Let contamination be random with item propensities q_i subject to $(1/n)\Sigma q_i = \pi$ and a Rosenbaum-style odds constraint $\mathrm{odds}(q_i)/\mathrm{odds}(q_j) \leq \Gamma^2\_\mathrm{sel}$. The worst-case expected bias is attained by a two-point propensity vector assigning the high value to the largest capacities; it equals $\pi \cdot \bar{d}$ at $\Gamma\_\mathrm{sel} = 1$, recovering random contamination, converges to the Proposition 1 top-k bound as $\Gamma\_\mathrm{sel} \to \infty$, and is continuous and nondecreasing in between. At reference settings the endpoints differ by roughly a factor of 1.5, so a defensible finite selection strength buys real tightening. Rather than an arbitrary worst case, then, the adversarial bound sits at the endpoint of a family with forty years of applied precedent.

5. The contamination robustness value

Bounds on a single model's clean score are useful, but leaderboards make comparative claims, and a comparison can survive uncertainty that would swamp either score alone. Take $A \succ B$ with observed margin $\hat{\Delta} = \hat{\mu}\_A - \hat{\mu}\_B > 0$. Under the two-sided model the worst case is unambiguous: A's score is deflated by at most $B\_A(\Lambda, \pi)$ and B's inflated by at most $U\_B(\rho\Lambda, \pi)$, where U is the mirror image of B built from per-item inflation capacities. Proposition 2 states that the claim holds across the entire identified set if and only if

$$\hat{\Delta} > B\_A(\Lambda, \pi) + U\_B(\rho\Lambda, \pi).$$

The extremal configuration is feasible because contamination assignments are unconstrained across models, so this is an equality of worst cases rather than a loose sufficient condition.

Everything the reader needs compresses into one number. Define the contamination robustness value

$$\Gamma^*(\pi) = \inf\{\Lambda \in [0,1] : B\_A(\Lambda, \pi) + U\_B(\rho\Lambda, \pi) \geq \hat{\Delta}\},$$

with $\Gamma^* = \infty$ when the claim survives even complete memorisation within the budget. Both bias maps are continuous and nondecreasing in Λ, so the infimum is well defined, and bisection finds it. In the simple regime the closed form is the contamination frontier $\pi\, \Lambda\, (1 + \rho) = \hat{\Delta}$, which makes the trade-off between how much contamination there is and how strong it is explicit.

Γ^* is to contamination what the E-value [9] is to unmeasured confounding: one number per claim, reportable next to the margin and interpretable without reading the paper that produced it. A claim with $\Gamma^* = 0.03$ tolerates a channel closing three percent of headroom on a tenth of the items; one with $\Gamma^* = 0.5$ tolerates a channel closing half. Computing Γ^* requires only the published scores, without a corpus, a reference model, or a reference benchmark, which is what lets the method reach closed-weight models and legacy claims. Interpreting it requires a view about plausible contamination strength; Section 9 supplies a shared, pre-frozen calibration table, and an analyst who prefers not to commit can report the whole $\Gamma^*(\Lambda, \pi)$ curve instead.

Per-model clean-score intervals use the Imbens–Manski critical value [31], which interpolates between the width appropriate for a point-identified parameter and that for a set, with set-level intervals as the conservative alternative. Corpus certificates use Benjamini–Hochberg control [32], with Benjamini–Yekutieli [33] as the dependence-robust sensitivity.

6. Estimation and inference

The bound $\hat{B}(\Lambda)$ is a trimmed upper mean of the per-item capacities, which places it in the well-understood class of L-statistics, and Theorem 4 exploits that. Under iid item sampling, continuity of the capacity distribution with positive density at its $(1-\pi)$-quantile, a positive population margin, and a strictly increasing bias map at the crossing point: $\hat{\Gamma}^*$ is strongly consistent; $\sqrt{n}(\hat{\Gamma}^* - \Gamma^*)$ converges to a normal law, with asymptotic variance from the delta method with slope $1/B'(\Gamma^*)$; and the paired nonparametric item bootstrap consistently estimates that law. Pairing matters here, because the two models are scored on the same items, so $\hat{\Delta}$ and $\hat{B}$ are dependent and only joint resampling reproduces their joint distribution.

Theory of this kind is worth only as much as its finite-sample behaviour, so we checked the rate directly. Across n from 100 to 25,600 the root-mean-square error of $\hat{\Gamma}^*$ declines with empirical slope −0.518 against the theoretical −0.5, with bias at or below 0.0025 throughout.

A subtler issue arises when per-item scores come from finitely many stochastic draws. With m draws per item the plug-in knapsack carries two opposing biases. Capacity is the minimum of two affine functions and therefore concave in the score, so Jensen's inequality pushes the estimate down, while selecting the top $\lceil \pi n \rceil$ capacities from noisy estimates pushes it up. Neither dominates in general, so we measured the net: −40% at m = 2, crossing to +4% at m = 10, and below 1% by m = 50. Joint sampling-and-identification intervals keep at least 99.3% coverage throughout, and the simple bound, which depends on the draws only through the unbiased grand mean, achieves 100%. Our prescription is concrete: sharp bounds require at least ten published draws per item, and below that the library falls back to the simple bound.

Inference across a corpus needs multiplicity control, since sixteen models generate dozens of adjacent comparisons. Each claim receives a paired-bootstrap fragility p-value against the null that it is not robust at the reference sensitivity model, with the standard add-one correction. Benjamini–Hochberg across the corpus [32] then yields certificates at a controlled false discovery rate, valid under the positive dependence shared items induce, with Benjamini–Yekutieli [33] as the dependence-robust check.

7. Experimental design

The empirical programme has four parts, each testing an assumption the previous part had to make.

Emergent contamination at small scale. Injecting a lift according to formula A1 and then recovering it would be circular, so we train clean and contaminated twins and let the lift emerge from real memorisation. A fraction π of test items, with realised labels, is planted in the training corpus and repeated a chosen number of times; the twins are otherwise identical. Across the capacity spectrum we use logistic regression, a multilayer perceptron, and a random forest, giving 15 configurations over 6 seeds each. $\hat{\Lambda}$ is the 95th-percentile lift envelope divided by headroom, the empirical counterpart of the A1 cap.

Emergent contamination at language-model scale. The same design runs on a same-family pair, Qwen2.5-0.5B and Qwen2.5-1.5B, over 905 MMLU items with fp32 LoRA injection. The grid is complete: eight contaminated fractions $\pi \in \{0.05, \ldots, 0.5\}$ crossed with three exposure counts, dose $\in \{1, 4, 16\}$, for 24 configurations.

Corpus dilution. The concentrated grid exposes models to leaked items with nothing else in the training stream, measuring a ceiling rather than the pretraining regime. Dilution holds the model, π, and the per-item exposure count fixed while interleaving each leaked chunk with mix $\in \{0, 4, 20\}$ neutral wikitext chunks, so total training grows while leaked exposures do not. Real pretraining differs from the concentrated grid in exactly that direction. It is replicated over three seeds and a second model.

Application. Two exploratory case studies and one pre-registered confirmatory audit run on published benchmark results (Section 10). Throughout, coverage is assessed against known ground truth wherever it exists, every configuration is reported including those that failed, seeds are fixed, versions locked, and each number is traceable to its generating script.

8. Ground-truth validation with emergent contamination

8.1 Small-scale twins: the channel, and two repairs

The first question is whether A1 describes what memorisation actually does. It does: the 95th-percentile lift envelope tracks item headroom with correlation between +0.91 and +1.00 in all 15 configurations, so the headroom parametrisation describes the data rather than merely simplifying it.

Channel strength is stratified by capacity and dose, as the dose-response literature at language-model scale predicts [14]. Logistic regression gives $\hat{\Lambda}$ between 0.02 and 0.23, rising monotonically with dose; the multilayer perceptron reaches about 0.96 and the random forest 1.00. The $\Lambda = 1$ extreme is real rather than a theoretical limit: an interpolating learner that has seen an item answers it perfectly. Any calibration reporting one global Λ averages over a quantity varying by more than an order of magnitude.

Two assumptions failed, and both failures changed the model. Strict monotonicity does not hold: up to 27% of leaked items score lower in the contaminated twin, which is why U1 exists. Spillover is real at small scale: drift on clean items reaches 0.20 for the multilayer perceptron and broke naive coverage, which held in only 1 of 3 configurations; with the calibrated ε-widening of U4, coverage returns to 3 of 3 in every failing configuration. These are repairs rather than robustness checks, because the model without them is falsified by our own data.

Coverage under known parameters is otherwise as the theory requires. With injected contamination at known (Λ, π), interval coverage is 1.000 across all regimes, and drops to between 0.10 and 0.34 only when the analyst underestimates contamination by half. A method that never fails is not being tested; this one fails precisely when its input assumption is wrong, by an amount that tracks how wrong it is. A blind audit on a synthetic leaderboard with one secretly contaminated model flagged exactly that model's winning claim, at $\Gamma^* = 0.173$ against $\Gamma^* \geq 0.29$ for the clean claims.

8.2 Language-model scale: a dose law

The 24-configuration grid reveals a dose law the small-scale runs only hinted at. A single exposure is net-harmful: mean lift is negative in 7 of the 8 dose-1 configurations, reaching −2.9 points, with per-item violation rates between 31% and 56%. At dose 16 the picture inverts into total memorisation, with $\hat{\Lambda} = 1.000$ in all eight configurations and lifts as large as +56 points. The downward channel therefore describes what light contamination usually does, so U1 is not an optional safety margin. A one-sided model would misdescribe the most common regime in the wild.

The capacity comparison vindicates the headroom parametrisation. Raw lift anti-orders with capacity: the 0.5B model gains more percentage points than the 1.5B model, purely because it starts further from the ceiling. Expressing the same data as Λ restores the correct ordering, so raw clean-versus-dirty gaps, the quantity the contamination literature usually reports, are not comparable across models, and the headroom normalisation is what makes them so.

Spillover grows with dose, from 0.04 to 0.30, and is always positive, consistent with format familiarity. Three of the 24 cross-configuration consistency checks fail, and all three occur exactly where $\hat{\Lambda}$ was pooled across doses: direct evidence for dose-stratified calibration, obtained from the failure mode rather than from an argument.

8.3 Corpus dilution: two regimes, not one

The concentrated grid measures a ceiling; real pretraining embeds leaked items in a vastly larger corpus. Dilution turns out not to be one phenomenon.

At dose 1, a single exposure per item and the closest analogue of scraped pretraining, $\hat{\Lambda}$ at the 95th percentile falls from 0.58 through 0.38 to 0.35 as the mix increases, moving from above the calibrated field range [0.10, 0.45] to inside it. This is the calibration loop closing: the literature-derived range of Section 9 had been an external prior, and is here reproduced by direct interpolation in a controlled experiment. Single-shot contamination embedded in a corpus lands at field strength.

At dose 4 dilution runs the other way: $\hat{\Lambda}$ climbs from 0.73 to 0.98, lift triples, and per-item violations collapse from 0.18 to 0.03. Interleaving neutral text between repeated exposures is spaced rather than massed practice, and consolidates memorisation instead of washing it out. Deliberate or repeated contamination therefore saturates toward $\Lambda = 1$ however dilute the corpus.

Replication over three seeds and a second model separates the stable findings from the noisy. The consolidation endpoint replicates almost exactly: $\hat{\Lambda}$ at dose 4 and mix 20 is 0.976, 0.988, and 0.982 across seeds, a range of 0.012, and the dose-4 trend is monotone in every seed. The dose-1 diluted endpoint also replicates inside the field range, at 0.286 to 0.346. The dose-1 concentrated starting point does not, ranging from 0.27 to 0.58, so we claim the diluted endpoint and the consolidation law as replicated and label the concentrated dose-1 level noisy. The 0.5B model shows the same structure at lower levels, with dose-1 lift near zero and dose-4 consolidation rising from 0.68 to 0.76, consistent with the capacity law.

The field range is therefore validated for accidental, scrape-style contamination. Section 9's concentrated measured-ceiling policy is also necessary, because repeated-exposure contamination exceeds even that ceiling. No single constant covers both regimes, so calibration must be stratified by dose, as the pre-registration required for reasons that were then theoretical and are now measured.

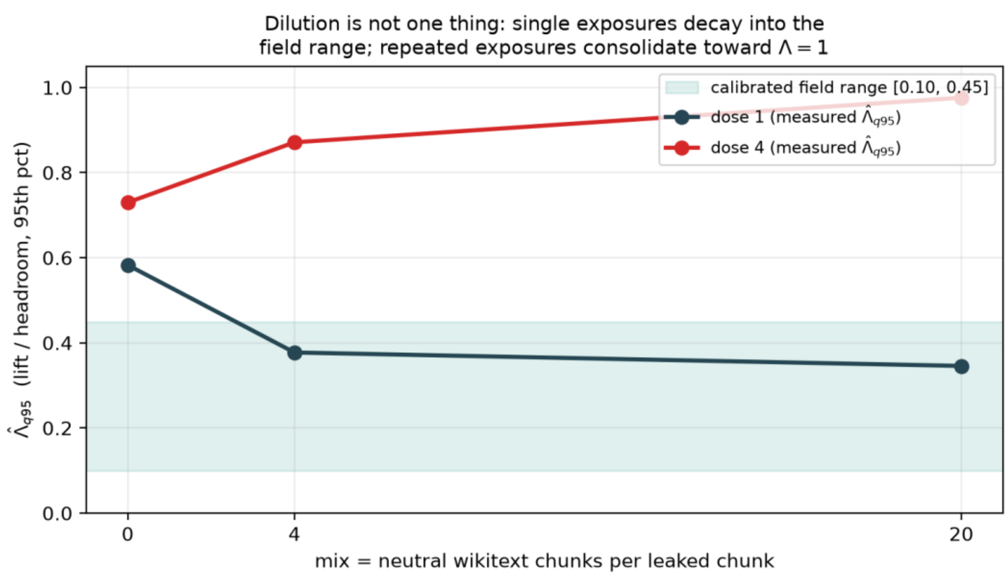

Figure 2: Corpus dilution behaves differently at different doses. Each leaked chunk is interleaved with mix ∈ {0, 4, 20} neutral wikitext chunks, holding per-item exposures fixed while total training grows. A single exposure (dose 1) decays into the calibrated field range [0.10, 0.45], the regime describing accidental scraping. Repeated exposures (dose 4) consolidate toward complete memorization ($\hat{\Lambda} = 0.98$), the regime describing deliberate or recurring contamination. Session-reproduction noise on $\hat{\Lambda}$ is at most 0.039, so the separation between the curves far exceeds the measurement error.

9. Calibrating contamination strength from published evidence

Interpreting Γ^* requires a view about how strong contamination actually is, and the published literature already contains the measurements needed to form one. Every clean-versus-dirty comparison converts into our units by dividing the reported lift by the headroom it had to work with. The LLaMA-70B HellaSwag result, a 15.3-point gap on 36.5 points of headroom [1], gives $\Lambda \approx 0.42$. GPT-3 on SQuAD [34] gives about 0.40. The Llama 3 flagged-subset estimates [2] give 0.40 to 0.47, and are upper bounds by construction, since flagging selects the items most likely to have leaked. A clean mirror of GSM8K gives about 0.21, C-Eval about 0.31, and MMLU about 0.02.

The pattern is strongly heterogeneous across tasks, which is itself informative: contamination strength depends on how memorisable a task's items are, not only on how much leaked. Pooling gives a meta-analytic central range of [0.1, 0.45]. Our calibration table holds eighteen sourced rows, eight of them quality-eligible for the pooled median under the pre-registered policy; rows from our own measurements enter only as upper-bound anchors, so the audit is not calibrated against its own experiments. Λ is stratified by model, following the dose-response law [14], and estimates from ConStat [16] and independent contamination reports [35] enter on the same footing. In the confirmatory audit we interpret every Γ^* against this range, frozen before any confirmatory number was computed.

10. Application to real benchmarks

10.1 Exploratory case studies

Time-series foundation models. The first case study uses a transfer atlas of 6 frozen models on 17 datasets with continuous MASE-skill scores. Continuous scores place it in regime R1, so the sharp knapsack applies, and the corpus supplies its own contamination-prior annotations as provenance strata. Every adjacent ranking claim is fragile: $\Gamma^* \leq 0.114$ at $\pi = 0.1$, at or below the bottom of the calibrated range. Stratification cannot rescue margins of 1.3 skill points or less, because 14 of the 17 datasets carry a high contamination prior. This agrees with the emerging contamination-auditing literature in this domain [4], [19], reached without any access to pretraining corpora.

Open LLM Leaderboard, ARC-Challenge. The second case study uses 1,170 paired items across 4 models with single-draw binary scores, routed to the simple regime, with 500 bootstrap replicates. All three adjacent claims separate cleanly. Llama-3-8B $\succ$ Mistral-7B, at a margin of 0.26 points, has $\Gamma^* = 0.026$: a whisker of contamination flips it. Mistral-7B $\succ$ Llama-2-13B, at 1.9 points, has $\Gamma^* = 0.188$, inside the calibrated range and therefore genuinely in doubt. Llama-2-13B $\succ$ Llama-2-7B, at 5.0 points, has $\Gamma^* = 0.504$, above the entire range, and earns a false-discovery-controlled certificate at $p = 0.006$. Independent scaling evidence says precisely that claim should survive.

10.2 The pre-registered confirmatory audit

Everything above is exploratory. The confirmatory analysis was frozen in advance: protocol version 1.0, dated 19 July 2026, at git tag prereg-freeze-v1.0, registered before any confirmatory Γ^* was computed. It specifies 16 fixed candidate models across 5 Open LLM Leaderboard v1-archive tasks, adjacent-pair claims only, a primary rule of $\Gamma^*(\pi = 0.1) < \Lambda\_ref$ with Λ_ref pooled at 0.355 and 0.42 for the HellaSwag stratum, Benjamini–Hochberg as primary with Benjamini–Yekutieli as sensitivity, and $\rho \in \{0, 0.2\}$. Three deviations were logged mechanically at run time: TruthfulQA was dropped because the archive stores mc1 and mc2 rather than the frozen acc and acc_norm specification; Falcon-7B was dropped on ARC because its snapshot shares no items with the modal item universe; and GPT-J was dropped on Winogrande for a malformed score column.

The counts come out lopsided. Of 58 adjacent claims, 49 (84.5%) are not contamination-robust, with an exact two-sided 95% interval of [72.6%, 92.7%] and a one-sided p-value of 2.7×10^{-21} against the pre-registered 25% threshold. Excluding pairs overlapping the pilot leaves 47 of 56, or 83.9%, interval [71.7%, 92.4%], and the ρ

= 0.2 sensitivity leaves the count unchanged. Per task, the median Γ*(0.1) is between 0.084 and 0.120 on HellaSwag, Winogrande, and ARC, so the typical adjacent claim tolerates less than a third of the calibrated contamination strength; on GSM8K, whose margins are larger, it is 0.212. Only 8 claims earn Benjamini–Hochberg certificates and 6 survive Benjamini–Yekutieli. Every certified claim has a margin above 5 points, and they are of two kinds: cross-generation capability gaps, such as Llama-3-70B over Qwen1.5-14B on GSM8K, and comparisons in which everything beats GPT-J. The leaderboard's ordering of eras is contamination-robust; its ordering of neighbours is not.

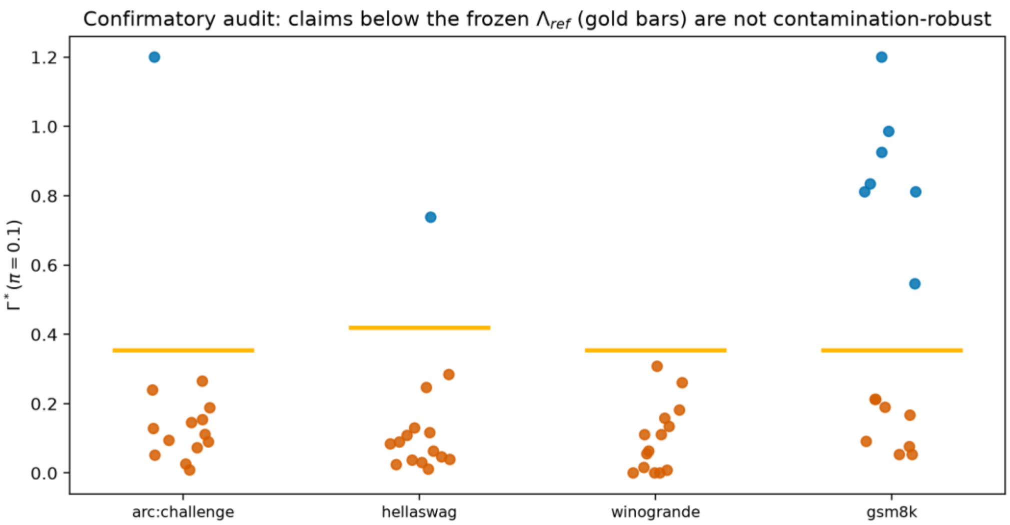

Figure 3: The pre-registered confirmatory audit. Each point is one adjacent leaderboard claim's contamination robustness value Γ* at π = 0.1; gold bars mark the frozen reference strength Λ_ref (0.42 on HellaSwag, 0.355 elsewhere). Vermillion points fall below the calibrated contamination strength and are not robust (49 of 58, or 84.5%); blue points survive. Certified claims cluster at large margins, and none has a margin below 5 points.

The exact test treats claims as independent, and adjacent claims are not: consecutive pairs share a model, and the same sixteen models recur across tasks. Three checks show the conclusion does not lean on that assumption. Splitting each task's chain into alternating halves, which share no model within a task, gives 83.3% and 85.7% non-robust, each at $p \approx 3\times10^{-11}$. Every task rejects the pre-registered threshold on its own, the weakest being GSM8K at $p = 0.017$. And at the observed rate the exact test rejects for any effective sample size of three or more claims, so dependence would have to destroy nearly all the information in the corpus to change the answer.

The protocol was registered on the Open Science Framework (https://osf.io/2436z, DOI 10.17605/OSF.IO/2436Z, 20 July 2026, public and un-embargoed) before any confirmatory number was publicised. The registration carries the protocol verbatim, the frozen calibration table, and a chain-of-custody addendum recording which files were frozen at tag prereg-freeze-v1.0 (commit ca7773b) and which were written afterwards, including the driver script, the item-universe amendment, and a post-freeze correction to a confidence-interval report that left every estimate unchanged. A mirror archived at the Internet Archive provides a second timestamp not under author control.

10.3 What the audit reduces to, and why that is the result

A reader who works through Section 5 will notice something the headline conceals, so we state it ourselves. In the single-draw binary regime the simple bound gives $\Gamma^*(\pi) = \hat{\Delta} / (\pi(1 + \rho))$, so the pre-registered rule "not robust if and only if $\Gamma^*(0.1) < \Lambda\_ref$" is algebraically identical to

$$\hat{\Delta} < \pi \cdot \Lambda\_ref = 0.0355 \quad (0.042 \text{ on HellaSwag}),$$

a threshold on the raw margin. We verified that this equivalence holds for all 58 claims. The confirmatory audit could have been executed with a ruler.

We regard this as the result rather than as a defect, and three things follow from it.

First, the arithmetic was never the hard part. Anyone can declare that gaps below three and a half points are untrustworthy; which threshold is defensible is a measurement problem, not an arithmetic one. Ours is π times the median of eight quality-filtered estimates of contamination-induced lift, converted to a common headroom parametrisation (Section 9), frozen before unblinding, and attached to a third-party timestamp. What we contribute is the derivation of 0.0355 from contamination evidence; comparing a margin against it is arithmetic.

Second, the reduction is a property of the data regime rather than of the method. Γ^* collapses to $\hat{\Delta}/\pi$ only when items are binary and measured once, because the per-item deflation capacity degenerates (Lemma A.1). That is a statement about how impoverished single-draw accuracy data is, and it is now a theorem rather than an observation. Theorem 5 shows that with one binary draw per item the observable law is Bernoulli in the mean alone, that two exactly indistinguishable populations exist whose clean means differ by precisely $\pi\Lambda$, and therefore that no procedure can certify robustness beyond $\hat{\Delta}/(\pi(1 + \rho))$ from such data. The ruler is minimax-optimal. Where richer data exists the machinery separates: with continuous per-item scores the sharp knapsack binds strictly tighter than $\pi\Lambda$, as the time-series atlas shows, and at 20 repeated draws it certifies $\Gamma^* = 0.172$ against the simple bound's 0.154. Leaderboards audited here publish the weakest data they could, and the reduction is what that choice costs. It is also the argument for Section 11's prescription.

Third, the reduction makes the result auditable: a sceptical reader can re-derive every classification from the published margins without trusting our code, and can see at once what would change it.

Sensitivity to the frozen constant. The decision depends on π and Λ only through their product, so the whole plane collapses to one curve, which we sweep. Non-robust claims make up 44.8% of the corpus at $\Lambda = 0.10$, 70.7% at 0.20, 82.8% at 0.30, and 84.5% at the frozen 0.355, staying flat at 84.5% throughout $\Lambda \in [0.355, 0.50]$. The pre-registered hypothesis clears its 25% bar for every $\Lambda \geq 0.055$ and is significant at $\alpha = 0.05$ for every $\Lambda \geq 0.080$. For the headline to fail, true contamination strength would have to sit below about 0.055: beneath the floor of the calibrated field range [0.10, 0.45], below seven of the eight quality-eligible estimates, and an order of magnitude below the concentrated-exposure ceiling measured in Section 8. The qualitative finding survives any calibration a reviewer could reasonably substitute; only the exact percentage moves.

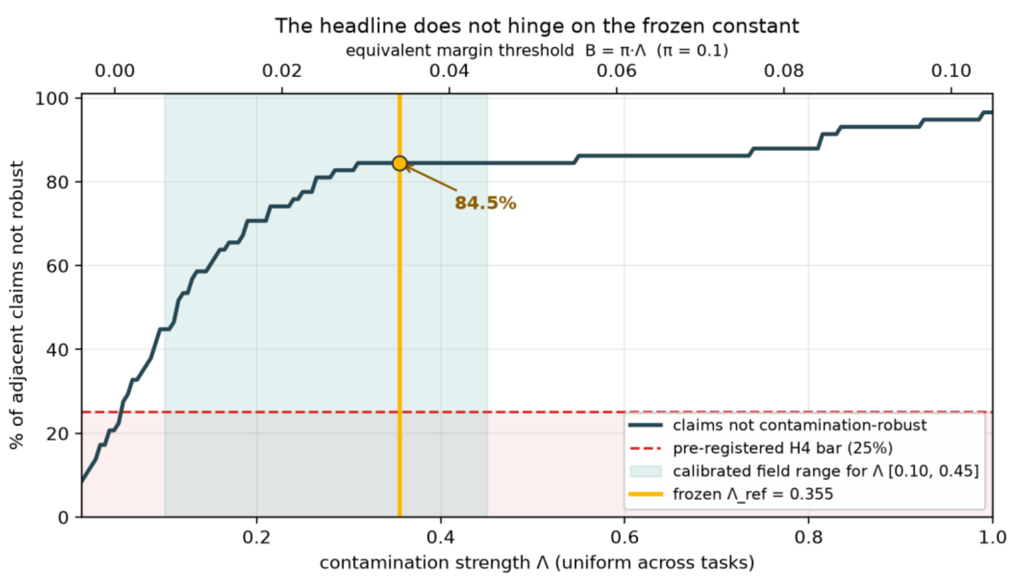

Figure 4: The headline does not hinge on the frozen constant. Because the decision rule depends on (π, Λ) only through the product B = π·Λ, the parameter plane collapses to one curve, swept here across the full plausible range. The pre-registered hypothesis clears its 25% bar for every $\Lambda \geq 0.055$, below the floor of the calibrated field range (shaded), and the non-robust fraction is flat above the frozen value.

10.4 Is the headline an artifact of auditing adjacent pairs?

Adjacent pairs are selected to have the smallest margins in a ranking, so the objection writes itself: 84.5% may be produced by the claim-selection rule rather than by contamination sensitivity. We took it seriously enough to measure. Applying the identical frozen rule to all 450 model pairs across the four tasks, the full comparison set with no selection at all, gives 30.4% non-robust. Stratifying by rank gap gives Table 1.

rank gap	non-robust	median margin
1 (adjacent)	49/58 = 84.5%	0.011
2	39/54 = 72.2%	0.023
3	28/50 = 56.0%	0.033
4	16/46 = 34.8%	0.046
5	5/42 = 11.9%	0.054
≥ 6	0/102 = 0.0%	≥ 0.068

Table 1: Contamination robustness as a function of rank distance, under the frozen decision rule applied without selection to all 450 model pairs. The non-robust proportion falls monotonically as the two models move further apart, and vanishes beyond a gap of five positions. The right-hand column shows the gradient is driven by margin: adjacent pairs differ by about one point, pairs six or more positions apart by seven or more. Since no selection was applied, the table is also the answer to the selection objection.

The objection is correct that adjacency selects small margins, and the gradient this produces is the cleanest quantitative statement of our thesis in the paper. Robustness varies systematically with rank distance: it is a monotone function of how far apart two models sit, and comparisons across five or more rank positions are essentially all robust — 2.1% non-robust — while comparisons between neighbours are essentially all fragile.

We report both numbers and say what each supports. The 30.4% all-pairs figure answers how often contamination sensitivity matters across arbitrary comparisons. The 84.5% adjacent figure answers the question a leaderboard actually poses, because a leaderboard is an ordering: it invites the reader to compare entry k with entry k+1, and those are exactly the comparisons that do not survive. Neither number carries the headline alone. The gradient running between them does.

What would change our mind. The finding is falsifiable on three fronts, all pre-specified in the registration. If contamination-induced lift in the wild were measured below $\Lambda \approx 0.055$ for these benchmarks, the hypothesis would fail; that is one well-designed injection study away, and we would report it. If the bounded-headroom envelope A1 were rejected at calibrated Λ, the audit would be re-run under the full-memorisation channel and both results reported. If per-item records proved systematically unusable, the corpus would fall back to continuous-score data and the substitution would be disclosed. Each is a study someone could run against us.

11. Design sensitivity: what buys robustness

Because Γ* depends on the benchmark's design as well as the models' scores, builders can raise it deliberately. We quantify two levers and price both.

Freshness is the first. At a 3-point margin, raising the post-cutoff item fraction from 0 to 90% lifts Γ* from 0.168 to 0.453, moving a claim from fragile across the whole calibrated range to robust across it. Stratification U3 supplies the mechanism. Post-cutoff items are clean by construction, so the adversary's budget cannot be spent on them. Dating benchmark items buys identification, and it costs almost nothing.

Repeated draws are the second. At m = 20 draws per item the sharp bound certifies $\Gamma^* = 0.172$ against the simple bound's 0.154, roughly 12% more certified robustness, and below m = 10 it is unreliable for the finite-draw reasons of Section 6. Publishing at least ten draws per item entitles a benchmark to sharp identification; publishing a single binary draw forfeits it, and Theorem 5 says the forfeit is irreversible by any amount of statistical cleverness.

12. Discussion

Why does this work at all? Because contamination, unlike most nuisance parameters in evaluation, has a physical mechanism reproducible in a laboratory. One can train a model on leaked items and watch what happens, which makes Λ a measured constant rather than a prior. That distinguishes this sensitivity analysis from its ancestors in epidemiology, where the strength of an unmeasured confounder can only be argued about.

The results also suggest a reading of leaderboards that is neither credulous nor nihilistic. Contamination does not make benchmarks useless; Table 1 shows it makes them coarse. A leaderboard reliably separates models that differ by a generation and unreliably separates models that differ by a point, and the boundary between those claims is now computable rather than intuited.

Generalisation beyond the settings audited here is a matter of the calibration table rather than of the method. Γ^* is computable for any benchmark that publishes per-item or aggregate scores; what changes across domains is the plausible range of Λ, and Section 9's table is meant to be extended by anyone who runs an injection study. Accidental and deliberate contamination occupy different parts of that range, and pooling them produced the only consistency failures we observed. Practically, the proposal costs one column: a leaderboard reporting Γ^* beside each margin lets readers see which orderings would survive the contamination everyone already suspects, without anyone having to prove it occurred.

13. Limitations and threats to validity

Scale is the main gap. Axiom A1's channel shape is validated with real memorisation at small scale, at language-model scale, and under corpus dilution, but the largest injection model here is 1.5B parameters, and the dilution sweep covers two models, one contaminated fraction, and three seeds. Seed noise is small where it matters, 0.012 at the consolidation endpoint, and large at concentrated single exposure, where it reaches 0.31; the conclusions quoted here are the seed-stable ones. Whether the dose-1 interpolation holds at 70B-scale pretraining is an extrapolation we flag rather than a measurement we own.

In the single-draw binary regime, which covers most current leaderboards, Λ and π enter only through their product, so the frontier earns its second dimension only under repeated draws or provenance strata. We state this plainly, prove it is a property of the data rather than of the estimator (Theorem 5), and offer Section 11 as the constructive response.

Calibrating Λ from pre-cutoff and post-cutoff comparisons can be confounded by temporal drift, since later items may differ in difficulty as well as cleanliness; the pre-registration pairs items on difficulty and reports calibration uncertainty. Both case studies are exploratory. The spillover term ε should vanish at realistic corpus scale, but we measure it rather than assume it away.

Finally, independence of verification. This artifact was audited by its own author across four forensic iterations, one of which added a critical guard for the binary regime; an independent adversarial audit is the correct next scrutiny and has not happened. Our repository carries a verification log recording every attempt to re-run the artifact, each entry labelled by the runner's independence; it currently holds two. The first is author-level and established only that the replication instructions execute from a clean clone, after a step that did not was fixed. The second is coauthor-level: a second author cloned the public repository and re-derived every result file byte-identically, all steps passing. Both ran on the same machine and interpreter, so neither speaks to cross-platform reproduction, and neither is independent in the sense that matters. Readers can see the state of verification rather than infer it, and discrepancy reports of any size are solicited.

Formal statements and proofs of Propositions 1 to 3, Theorems 4 and 5, and the supporting lemmas (R^2 degeneracy, finite-draw bias directions, Imbens–Manski properties, and the spillover extension) are in Appendix A, each pinned to a named unit test.

14. Conclusion

For a closed-weight model the question "is this benchmark contaminated?" has no answer, and forty years of causal inference say the right response to an unanswerable question is to change it. This paper asked instead how much contamination it would take to overturn a given claim, and built the chain needed to answer that: a sensitivity model whose axioms were tested against real memorisation and twice repaired when they failed; sharp bounds with estimation theory and a minimax-optimality result for the data regime leaderboards actually publish; a calibration of contamination strength that a corpus-dilution experiment reproduces in the laboratory for the scraped-pretraining case; and a pre-registered audit whose every constant was frozen and timestamped before unblinding.

The empirical picture is specific rather than sweeping. Robustness on the Open LLM Leaderboard is a monotone function of rank distance: comparisons across five or more positions essentially all survive, while 84.5% of neighbouring claims do not. A leaderboard is trustworthy about which era of models beats which, and untrustworthy about the pairwise comparisons its ranking format invites. One extra column fixes it: report Γ^* next to every margin. Benchmarks that want more than the simple bound can buy it, since dated items and published repeated draws give sharper identification. Sensitivity analysis will not tell anyone whether a benchmark is clean; it will tell every reader which conclusions would survive if it were not, which is the question they were asking all along.

15. Reproducibility statement

Code, data artifacts, and this manuscript are public at https://github.com/tanviranjumjoarder/contamsens under an MIT licence. Seed 42 is used throughout with an exact-version lock, and 57 unit tests include a constructive test of sharpness attainment. One command regenerates every fast number in under two minutes; a full flag adds the twin-training grid in roughly thirty minutes on CPU. Every reported number maps to its generating script in the provenance file, and the package installs as a standard editable Python package (v0.2.0).

The confirmatory analysis of Section 10.2 was pre-registered at https://osf.io/2436z (DOI 10.17605/OSF.IO/2436Z, 20 July 2026), with the analysis library and calibration constants frozen beforehand at git tag prereg-freeze-v1.0 (commit ca7773b), publicly resolvable in the repository above, so the freeze is independently checkable rather than asserted. The pre-registration, theory notes, and calibration table in the repository are byte-identical to that tag and to the copies attached to the registration, and deviations are enumerated in the chain-of-custody addendum filed with it.

Appendix A — Proofs

Every result below is implemented in the accompanying library and pinned by a unit test, named in brackets after each proof.

A.1 Setup and notation

Items are indexed $i = 1, \ldots, n$. For a fixed model, the observed scores are $y = (y_1, \ldots, y_n) \in [0,1]^n$, the clean (counterfactual) scores are $y^* \in [0,1]^n$, and the latent contamination indicators are $c \in \{0,1\}^n$. The estimand is $\theta = n^{-1}\Sigma_i\, y^*_i$ and the observed mean is $\hat{\mu} = n^{-1}\Sigma_i\, y_i$.

CSM(Λ, π), with $\Lambda, \pi \in [0,1]$, consists of (A1) $0 \le y_i - y^*_i \le c_i \Lambda (1 - y^*_i)$ for all i, and (A2) $n^{-1}\Sigma_i c_i \le \pi$.

For $\Lambda < 1$ define the floor $\ell_i(\Lambda) = \max\{0, (y_i - \Lambda)/(1 - \Lambda)\}$ and the deflation capacity $d_i(\Lambda) = y_i - \ell_i(\Lambda)$; for $\Lambda = 1$ set $\ell_i = 0$ and $d_i = y_i$. Write $k = \lceil \pi n \rceil \wedge n$ and let $d_{(1)} \ge \ldots \ge d_{(n)}$ be the sorted capacities.

Lemma A.0 (capacity form). For $\Lambda \in (0,1)$, $d_i(\Lambda) = y_i$ if $y_i \le \Lambda$, and $d_i(\Lambda) = \Lambda(1 - y_i)/(1 - \Lambda)$ if $y_i > \Lambda$. Equivalently $d_i(\Lambda) = \min\{ y_i, \Lambda(1 - y_i)/(1 - \Lambda) \}$, so d_i is the minimum of two affine functions of y_i and is therefore concave in y_i. It is continuous, is maximised at $y_i = \Lambda$ with value Λ, and is nondecreasing in Λ.

Proof. Solving (A1) for y^*_i with $c_i = 1$ gives $y_i \le \Lambda + (1 - \Lambda)y^*_i$, that is $y^*_i \ge (y_i - \Lambda)/(1 - \Lambda)$; intersecting with $y^*_i \ge 0$ gives the floor. The minimum form follows by checking which branch binds at $y_i \lesseqgtr \Lambda$, and the derivative of the right branch in Λ is $(1 - y_i)/(1 - \Lambda)^2 \ge 0$. ■ [test_deflation_hand_computed, test_deflation_peak_at_lambda]

A.2 Proposition 1 (sharp identified set)

Statement. Under CSM(Λ, π), the identified set for θ given y is exactly the interval $\Theta(y) = [\hat{\mu} - B(\Lambda, \pi), \hat{\mu}]$, where $B(\Lambda, \pi) = n^{-1} \Sigma_{j=1}^{k} d_{(j)}(\Lambda)$.

Proof. ($\subseteq$) For any feasible (c, y*), $\theta = \hat{\mu} - n^{-1}\Sigma_{i: c_i=1}(y_i - y^*_i)$. Each term obeys $0 \le y_i - y^*_i \le d_i(\Lambda)$ by Lemma A.0, and at most k indices have $c_i = 1$, so the sum of any k or fewer capacities is at most the sum of the k largest. Hence $\theta \in [\hat{\mu} - B, \hat{\mu}]$. ($\supseteq$, sharpness) The upper endpoint is attained by $c \equiv 0$ and $y^* = y$. The lower endpoint is attained by setting $c_i = 1$ exactly on an argmax set S of k capacities, with $y^*_i = \ell_i(\Lambda)$ for $i \in S$ and $y^*_i = y_i$ otherwise; (A1) holds by construction of ℓ_i, since the constructed lift equals the cap. Intermediate values are attained by scaling y^*_i continuously between ℓ_i and y_i on S, θ being continuous in that scaling, so $\Theta(y)$ is the full interval. ■ [test_sharpness_attained, test_interval_contains_mu_and_is_ordered]

Corollary 1 (distribution-free width). $B(\Lambda, \pi) \le \pi\Lambda + \Lambda/n$, since $n^{-1} k \max_i d_i \le (\pi + 1/n)\Lambda$ by the maximum in Lemma A.0. The identified-set width is therefore at most $\pi\Lambda$ up to the ceiling-rounding term, and the bound is achieved when k items sit at $y_i = \Lambda$. [test_simple_bound_dominates_sharp]

Corollary 2 (reductions). $B(0, \pi) = B(\Lambda, 0) = 0$, so Θ collapses to $\{\hat{\mu}\}$: sampling-only analyses such as rank intervals are the $\Lambda = 0$ special case. [test_reduction_at_zero]

Lemma A.1 (R2 degeneracy). If $y_i \in \{0,1\}$ for all i and $\Lambda < 1$, then $d_i(\Lambda) = 0$ for all i and hence $B(\Lambda, \pi) = 0$, while $B(1, \pi) = n^{-1}\Sigma_{j\le k} y_{(j)}$. So $B(\cdot, \pi)$ is identically zero on [0,1) with a jump at $\Lambda = 1$. The per-item machinery is consequently vacuous on single-draw binary data, and the population simple bound, of width $\pi\Lambda$ at the probability level, must be used instead.

Proof. $d(0) = 0$ on the left branch, and $d(1) = 1\cdot(1 - 1)/(1 - \Lambda) = 0$ on the right branch; at $\Lambda = 1$, $d = y$. ■ [test_sharp_bound_degenerates_on_binary, test_ceiling_effect]

A.3 Proposition 2 (claim robustness) and Γ*

Statement (two-sided form). For models A and B evaluated on shared items with margin $\hat{\Delta} = \hat{\mu}_A - \hat{\mu}_B > 0$, and under CSM±($\Lambda$, $\rho\Lambda$, π) applied to both models, the claim "A ≻ B holds for the clean scores" is true over the entire identified set if and only if $\hat{\Delta} > B_A(\Lambda, \pi) + U_B(\rho\Lambda, \pi)$, where U is the inflation analogue of B built from $u_i(\Lambda^-) = \min\{1 - y_i, y_i\Lambda^-/(1 - \Lambda^-)\}$.

Proof. $\theta_A \ge \hat{\mu}_A - B_A$ and $\theta_B \le \hat{\mu}_B + U_B$, with both endpoints attained by Proposition 1 and its mirror image applied to −y. The extremal joint configuration is feasible because A's and B's contamination assignments are unconstrained across models, so $\min(\theta_A - \theta_B) = \hat{\Delta} - B_A - U_B$. ■ [test_gamma_star_bisection_consistency, test_is_robust_rho_consistent_with_gamma_star]

Definition (Γ*). $\Gamma^*(\pi) = \inf\{\Lambda \in [0,1] : B_A(\Lambda,\pi) + U_B(\rho\Lambda,\pi) \ge \hat{\Delta}\}$, with $\Gamma^* = \infty$ if no such Λ exists. It is well defined by the continuity and monotonicity of both maps in Λ (Lemma A.0 and its mirror). In the simple regime the closed form is $\Gamma^* = \hat{\Delta} / (\pi(1 + \rho))$. [test_gamma_star_simple_closed_form, test_gamma_star_twosided_is_smaller]

A.4 Proposition 3 (selection-bounded bridge)

Statement. Let contamination be random with propensities $q_i = P(c_i = 1)$ subject to (i) $n^{-1}\Sigma q_i = \pi$ and (ii) $\text{odds}(q_i)/\text{odds}(q_j) \leq \Gamma^2\_\text{sel}$ for all i, j. Then the worst-case expected bias $B(\Lambda, \pi, \Gamma\_\text{sel}) = \max\_q\ n^{-1}\Sigma_i\, q_i\, d_i(\Lambda)$ satisfies: (a) it is attained by a two-point propensity vector taking values {q_lo, q_hi} with odds(q_hi) = $\Gamma^2_\text{sel}\cdot$odds(q_lo), the high value assigned to the t largest capacities for some t; (b) $B(\Lambda, \pi, 1) = \pi\, \bar{d}$, the random-contamination case, and $B(\Lambda, \pi, \infty)$ equals the Proposition 1 top-k bound; (c) B is nondecreasing and continuous in Γ_sel.

Proof. (a) The objective is linear in q over the polytope defined by (i) and (ii). Extreme points of that polytope have coordinates taking at most two distinct values, with the odds constraint active between them: any three distinct values admit a feasible transfer of mass toward larger d_i that increases the objective. Given the two values, assigning q_hi to the largest capacities is optimal by rearrangement. (b) At Γ_sel = 1 the only feasible q is the constant π. As Γ_sel → ∞, the assignment q_hi → 1 on t = k items and q_lo → 0 elsewhere is feasible in the limit, recovering the deterministic knapsack. (c) Enlarging Γ_sel enlarges the feasible set, and continuity follows from continuity of the extreme-point solution in Γ_sel. ■ [test_random_endpoint, test_adversarial_endpoint, test_monotone_in_gamma_sel_and_bracketed]

A.5 Theorem 4 (consistency and rate of $\hat{\Gamma}^*$)

Statement. Let items be iid with per-item scores distributed as F (continuous case, regime R1), and let $d(\cdot; \Lambda)$ be as in Lemma A.0. Define $B(\Lambda)$ as the trimmed upper mean at level π, $B(\Lambda) = E[d\cdot 1\{d \geq q\_\{1-\pi\}(d)\}]$ plus the boundary term, and assume that F_d is continuous with positive density at its $(1 - \pi)$-quantile for the relevant Λ, that $\Delta = \mu\_A - \mu\_B > 0$, and that $B(\Gamma^*) = \Delta$ with B strictly increasing at $\Gamma^* \in (0,1)$. Then (i) $\hat{\Gamma}^* \to \Gamma^*$ almost surely; (ii) $\sqrt{n}(\hat{\Gamma}^* - \Gamma^*) \Rightarrow N(0, \sigma^2)$, where σ^2 is the variance functional of $(\hat{\Delta}, \hat{B})$ obtained by the delta method with slope $1/B'(\Gamma^*)$; and (iii) the paired nonparametric item bootstrap consistently estimates the law in (ii).

Proof sketch. $\hat{B}(\Lambda)$ is a trimmed L-statistic with a bounded, monotone weight generator. Under the quantile-density condition it is strongly consistent and Hadamard differentiable at F tangentially to the uniform ball, by arguments of the van der Vaart–Wellner type (Lemma 22.10 and its relatives), which gives a $\sqrt{n}$ central limit theorem jointly with $\hat{\Delta}$ — a sample mean on the same items, the pairing supplying the joint law. Pointwise almost-sure convergence of the nondecreasing functions $\hat{B}(\cdot)$ to the continuous limit $B(\cdot)$ upgrades to uniform convergence on [0,1] by Pólya's argument. The map $(B, \Delta) \mapsto B^{-1}(\Delta)$ is differentiable at (B, Δ) with $B'(\Gamma^*) > 0$, so the functional delta method yields (ii); Hadamard differentiability together with bootstrap consistency for L-statistics yields (iii). ■ Empirical verification: RMSE slope −0.518 against the theoretical −0.5 over $n \in [100, 25600]$, with bias at or below 0.0025.

A.6 Lemma A.2 (finite-draw bias directions)

With m Bernoulli draws per item, so that $\hat{p}_i \sim m^{-1}\text{Bin}(m, p_i)$: (i) by the concavity in Lemma A.0 and Jensen's inequality, $E[d(\hat{p}_i)] \leq d(p_i)$ pointwise, a downward smoothing bias; (ii) top-k selection over noisy capacities satisfies E[max-k mean of $d(\hat{p})$] ≥ max-k mean of $E[d(\hat{p})]$, an upward selection bias. The net sign is regime-dependent. Measured values run from −40% at m = 2 through +4% at m = 10 to below 1% at m = 50, which motivates the m ≥ 10 prescription and the simple-bound fallback, the latter depending on $\hat{p}$ only through the unbiased grand mean.

A.7 Lemma A.3 (Imbens–Manski critical value)

The equation $\Phi(c + w) - \Phi(-c) = 1 - \alpha$ has a unique root $c(w) \in [z\_\{1-\alpha\}, z\_\{1-\alpha/2\}]$ for every $w \geq 0$, and c is continuous and nonincreasing in w with $c(0) = z\_\{1-\alpha/2\}$ and $c(\infty) = z\_\{1-\alpha\}$.

Proof. The left-hand side is strictly increasing in c and increasing in w; evaluate at the bracket endpoints. □ The parameter-level interval $[\hat{\theta}\_lo - c\cdot\hat{\sigma}\_lo, \hat{\theta}\_hi + c\cdot\hat{\sigma}\_hi]$ therefore has asymptotic coverage of at least $1 - \alpha$ for every θ in the identified set, with equality at the endpoints. The conditions of Imbens and Manski (2004, Theorem 1) hold here: the endpoint estimators are jointly asymptotically normal, which Theorem 4 supplies, and the identified-set width is consistently estimated. [test_im_critical_value_limits, test_im_worst_case_parameter_coverage]

A.8 Spillover-robust interval

If in addition $|n^{-1}\Sigma\_\{i: c_i=0\}(y_i - y^*_i)| \leq \varepsilon$, a bound on the net drift of clean items, then every bound above extends by widening both endpoints by ε. The proof of Proposition 1 goes through with the decomposition $\theta = \hat{\mu} -$ (contaminated deflation) $-$ (clean-item drift). The quantity ε is estimable from twin experiments, and was measured between 0.003 and 0.30, growing with dose. [test_twosided_upper_moves_only_with_lam_minus]

A.9 Theorem 5 (minimax optimality of the simple bound in the binary regime)

Setting. Single-draw binary regime R2, under the iid item sampling of Theorem 4. Items are drawn from a population P over (p^*, c, δ); the analyst observes one draw $y \sim$ Bernoulli(p) per item, with $p = p^* + c\delta$, $E[c] \leq \pi$, and, under CSM±, $-\Lambda^- p^* \leq \delta \leq \Lambda^+(1 - p^*)$. The estimand is $\theta(P) = E[p^*]$; write $\mu(P) = E[p]$ and $\Lambda = \Lambda^+$.

Statement. (i) With one draw per item the observable law is iid Bernoulli($\mu(P)$), so populations with equal μ are exactly indistinguishable at every sample size. (ii) For interior μ, that is $\pi\Lambda \leq \mu \leq 1 - \pi(1 - \Lambda)$, the identified set for θ given μ is exactly $[\mu - \pi\Lambda, \mu]$ at $\rho = 0$, and the width $\pi\Lambda$ is attained by a feasible population. (iii) Any interval procedure uniformly valid over the model class has width at least $\pi\Lambda$, and any claim-robustness procedure can certify at most $\Gamma^* = \Delta/(\pi(1 + \rho))$.

Proof. (i) Conditional on an item, y is Bernoulli(p); marginalising over P, a single draw is Bernoulli(E[p]) and items are iid, so the sample is iid Bernoulli(μ). Higher moments of p are not identified from one draw per item. (ii) Upper endpoint: at $\rho = 0$ we have $\delta \geq 0$, so $\theta \leq \mu$, attained by the uncontaminated population $p^* \equiv \mu$. Lower endpoint: feasibility requires $\delta \leq \min(p, \Lambda(1 - p^*))$, and the two constraints cross at $p = \Lambda$, where $p^* = 0$ and $\delta = \Lambda$ are simultaneously feasible, so the per-item bias is at most Λ and $E[c\delta] \leq \pi\Lambda$. For attainment, let P_2 place mass π on contaminated items with $p^* = 0$ and $\delta = \Lambda$, hence $p = \Lambda$, and mass $1 - \pi$ on clean items with $p^* = (\mu - \pi\Lambda)/(1 - \pi)$, which lies in [0,1] exactly on the stated interior range. Then $\mu(P_2) = \mu$ and $\theta(P_2) = \mu - \pi\Lambda$, while the population P_1 with $p^* \equiv \mu$ and $c \equiv 0$ has $\mu(P_1) = \mu$ and $\theta(P_1) = \mu$. By (i) the two are indistinguishable, so both endpoints belong to the identified set, and by the bias bound nothing below $\mu - \pi\Lambda$ does. (iii) A uniformly valid interval must contain $\theta(P_1) = \mu$ and $\theta(P_2) = \mu - \pi\Lambda$ on data laws it cannot tell apart, so its width is at least $\pi\Lambda$. For a claim $A \succ B$ with margin Δ, applying the P_2 construction to model A at deflation Λ^+ and its mirror to model B at inflation $\Lambda^- = \rho\Lambda^+$ reverses the population margin whenever $\pi\Lambda(1 + \rho) > \Delta$ while leaving both observable laws unchanged. No procedure can therefore certify robustness at any $\Lambda > \Delta/(\pi(1 + \rho))$. ■ [test_minimax_indistinguishable_pair, test_minimax_no_narrower_certificate]

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